

PROJECT SNAPSHOT

1,470

EMPLOYEE RECORDS ANALYSED

32

ATTRIBUTES

16.12%

ATTRITION RATE

TOOLCHAIN

Python · Pandas · NumPy

Matplotlib · Seaborn

Tableau Public · Jupyter Notebook

DATA ANALYTICS · PROJECT

Employee Attrition Intelligence Dashboard

HR Analytics | Employee Retention Analysis

PRESENTED BY

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Problem Statement

WHAT IS EMPLOYEE ATTRITION?

Employee attrition refers to the gradual reduction of the workforce when employees voluntarily leave an organisation and are not immediately replaced. Persistent attrition disrupts business continuity, increases hiring cost, and lowers team morale.

WHY IT MATTERS

Replacing a single employee costs 1.5× to 2× their annual salary in recruitment, onboarding and lost productivity — making retention a strategic priority for HR.

PROJECT OBJECTIVES

#	OBJECTIVE
01	Identify the key factors that influence employee attrition
02	Measure attrition across departments, roles & demographics
03	Build an interactive Tableau dashboard for HR teams
04	Recommend data-driven retention strategies

SECTION 02

Dataset Overview

Source: IBM HR Analytics Employee Attrition & Performance Dataset

TOTAL RECORDS

1,470

employee observations

ATTRIBUTES

32

numerical + categorical features

TARGET VARIABLE

Attrition

binary — Yes / No

CLASS DISTRIBUTION

237 · 1,233

Left vs Stayed

KEY FEATURES STUDIED

CATEGORY	FEATURES	TYPE
Demographic	Age, Gender, Marital Status, Education, Distance From Home	Mixed
Job Profile	Department, Job Role, Job Level, Business Travel, Overtime	Categorical
Compensation	Monthly Income, Hourly Rate, Stock Option Level, Percent Salary Hike	Numerical
Experience	Total Working Years, Years at Company, Years in Current Role, Years Since Last Promotion	Numerical
Satisfaction	Job Satisfaction, Work-Life Balance, Environment Satisfaction, Relationship Satisfaction	Ordinal

SECTION 03

Data Preprocessing

A structured seven-step pipeline in Jupyter Notebook to prepare the dataset for analysis.

STEP

01 Import & Load

Loaded the CSV into a **Pandas DataFrame**; imported NumPy, Matplotlib and Seaborn for analysis.

STEP

02 Inspect Schema

Reviewed `df.info()`, dtypes and `df.describe()` to understand structure.

STEP

03 Missing Values

Verified via `isnull().sum()` — no missing entries across any column.

STEP

04 Duplicate Check

Applied `df.duplicated()` — dataset is unique, no duplicate employee IDs.

STEP

05 Outlier Detection

Used **Seaborn boxplots** on Monthly Income, Age & Years at Company; retained genuine tail values.

STEP

06 Data Cleaning

Dropped constant columns (*EmployeeCount*, *StandardHours*, *Over18*) and standardised label formatting.

STEP 07 • KEY

STEP Feature Engineering

Created **Attrition Flag** (Yes=1, No=0) for numerical aggregation, and derived **Attrition Rate** as an aggregate metric across segments for Tableau KPIs.

SECTION 04

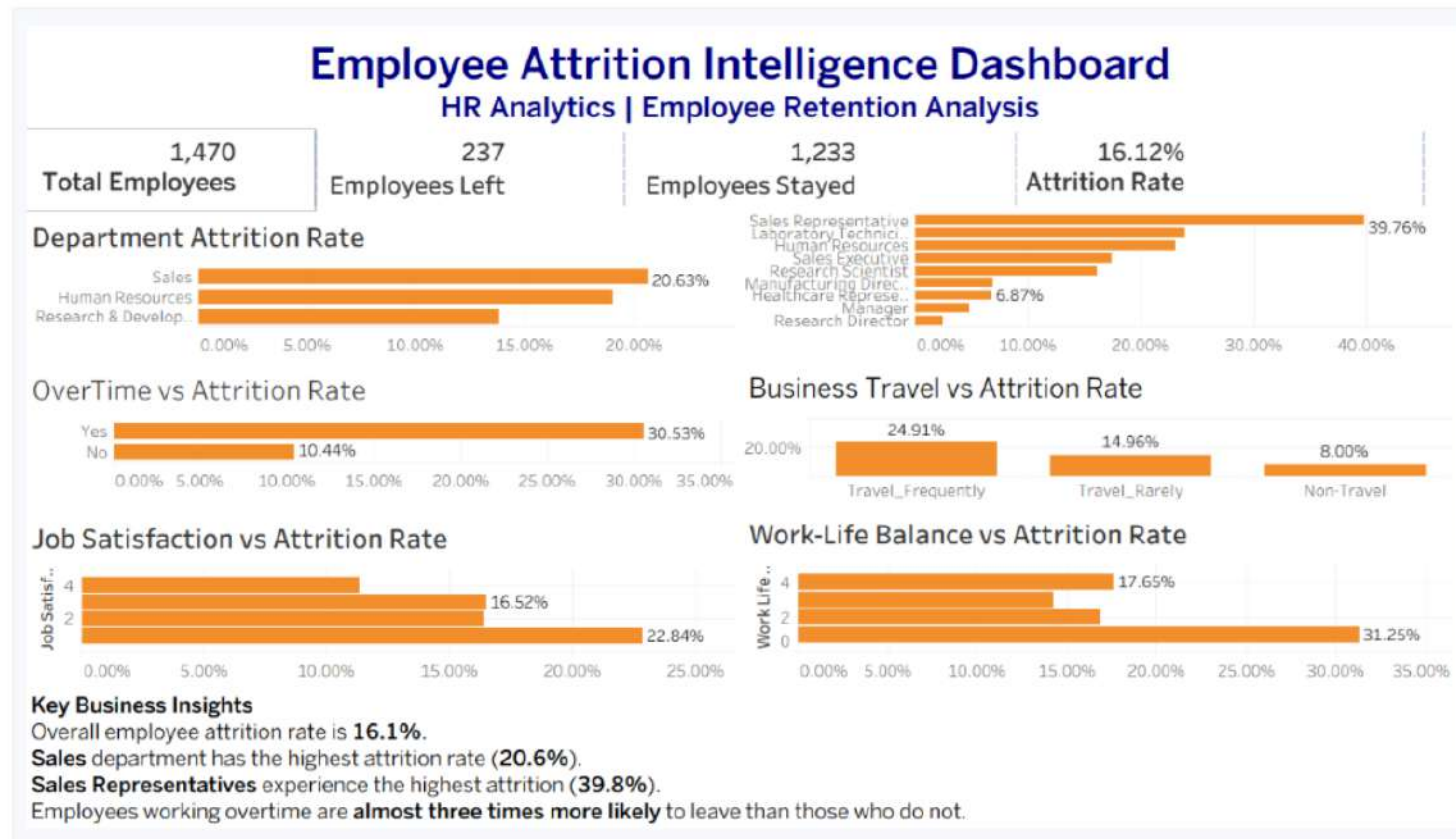
Exploratory Data Analysis

Nine core analyses performed in Python using Matplotlib & Seaborn, then reused in the Tableau dashboard.

#	ANALYSIS	CHART TYPE	OBSERVATION
01	Department-wise Attrition	Horizontal Bar	Sales leads at 20.63% — well above company average
02	Job Role-wise Attrition	Ranked Bar	Sales Representatives peak at 39.76%
03	Overtime vs Attrition	Grouped Bar	Overtime = Yes drives attrition to 30.53% vs 10.44%
04	Business Travel vs Attrition	Bar Chart	Frequent travellers churn 3× more than non-travellers
05	Monthly Income vs Attrition	Boxplot · KDE	Lower income brackets show higher attrition
06	Age vs Attrition	Histogram	Employees aged 18–30 exit most often
07	Job Satisfaction vs Attrition	Grouped Bar	Level 1 satisfaction shows 22.84% attrition
08	Work-Life Balance vs Attrition	Bar Chart	Rating 1 employees churn at 31.25%
09	Correlation Matrix	Seaborn Heatmap	Overtime, Marital Status and Job Level correlate strongest

SECTION 05

Tableau Dashboard



INTERACTIVE VIEW

HR analysts can filter by department, role & travel to drill into any attrition driver.

DASHBOARD COMPONENTS

- 01 **4 KPI Cards** — Total, Left, Stayed, Rate
- 02 **Department** attrition rate
- 03 **Job Role** ranked breakdown
- 04 **Overtime & Travel** comparison
- 05 **Satisfaction & Work-Life** impact
- 06 **Insight panel** with key takeaways

BUILT WITH

Tableau Public

Key Business Insights

Seven data-backed findings that HR should act on.

HEADLINE METRIC

16.1%

Overall employee attrition rate

Roughly 1 in every 6 employees left during the analysis period — a significant workforce loss that warrants targeted intervention.

237

EMPLOYEES LEFT

1,233

EMPLOYEES STAYED

01

SALES DEPARTMENT

20.6%

Highest attrition rate of any department

02

SALES REPRESENTATIVES

39.8%

Highest role-level attrition in the dataset

03

OVERTIME EFFECT

≈ 3x

Overtime workers are 3x more likely to leave

04

FREQUENT TRAVEL

24.9%

Attrition among frequent business travellers

05

LOW SATISFACTION & WLB

Rating 1 employees leave at up to 31.25% — poor work-life balance is a leading trigger.

06

YOUNGER WORKFORCE

Employees aged 18-30 leave far more often than mid-career staff.

SECTION 07

Recommendations

Six action areas HR can prioritise, mapped to the findings on the previous slide.

#	ACTION AREA	RECOMMENDED INITIATIVE	PRIORITY
01	Reduce Overtime Workload	Cap weekly overtime hours; redistribute workload & strengthen resource planning.	HIGH
02	Improve Work-Life Balance	Introduce flexible hours, remote-work windows and mandatory recovery days.	HIGH
03	Focus on Sales Retention	Redesign Sales Rep incentives, mentoring & onboarding pathway.	HIGH
04	Boost Engagement	Run pulse surveys, town-halls and manager one-on-ones every quarter.	MEDIUM
05	Career Growth Pathways	Publish clear promotion criteria, upskilling stipends and internal mobility.	MEDIUM
06	Recognition & Rewards	Formalise spot bonuses, peer-recognition and quarterly awards programmes.	MEDIUM

Expected outcome — reducing attrition even by 5 percentage points can save an organisation of this size an estimated ₹4–6 Cr annually in hiring & ramp-up costs.

SECTION 08

Conclusion

PROJECT SUMMARY

This project demonstrates how **data analytics** combined with an interactive **Tableau dashboard** can transform raw HR records into a decision-support tool.

Through Python-based EDA and visual storytelling, we identified the workforce segments most vulnerable to attrition — **Sales, overtime workers, frequent travellers and younger employees.**

The dashboard equips HR to move from reactive exit interviews to **proactive, data-driven retention.**

VALUE DELIVERED

- 01 Data-Driven Diagnosis**
Quantifies exactly which employee segments are at risk.
- 02 Interactive Exploration**
HR can filter and drill down without any code.
- 03 Actionable Recommendations**
Six prioritised initiatives mapped to observed drivers.

SCOPE FOR FUTURE WORK

Extend the pipeline with **machine-learning models** (Logistic Regression, Random Forest, XGBoost) to predict at-risk employees; integrate **real-time HR data** for a live monitoring layer.



Thank You

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